LAB | Create ELB to Route Traffic to Multiple Instances Extra

Introduction

In this hands-on lab, you will configure an Elastic Load Balancer (ELB) to route traffic to two EC2 instances. The ELB will distribute incoming requests to ensure high availability and scalability.

Objectives

By the end of this lab, you will be able to:

1. Create and configure security groups for ELB and EC2 instances.
2. Deploy and configure two EC2 instances running Apache (httpd) servers.
3. Set up an Application Load Balancer (ALB) and target group.
4. Test load balancing by accessing the ELB's DNS name.
5. Clean up all resources to prevent unnecessary costs.

Lab Instructions

1. Create a Security Group for EC2 Instances

- Navigate to the **AWS EC2 Console** → **Security Groups**.
- Create a new security group named `web-server-sg`.
- Configure **Inbound Rules**:
  - Allow **HTTP (port 80)** from anywhere (`0.0.0.0/0`).
  - Allow **HTTPS (port 443)** from anywhere (`0.0.0.0/0`).
  - Allow **SSH (port 22)** from your IP (for troubleshooting purposes).
- Click **Create Security Group**.

2. Launch Two EC2 Instances

- Navigate to the **AWS EC2 Console** → **Instances**.

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- Click **Launch Instance** and configure the following:
  - **Instance Name:** `Server1`.
  - **AMI:** Amazon Linux 2.
  - **Instance Type:** `t2.micro`.
  - **Security Group:** Select `web-server-sg`.
  - **User Data:** Use the following script to install and configure Apache:

```
#!/bin/bash
sudo yum update -y
sudo yum install -y httpd
echo "<h1>This is Server 1</h1>" | sudo tee /var/www/html/index.html
sudo systemctl start httpd
sudo systemctl enable httpd
```

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  - Click **Launch Instance**.
- Repeat the process for `Server2`, modifying the User Data:

```
#!/bin/bash
sudo yum update -y
sudo yum install -y httpd
echo "<h1>This is Server 2</h1>" | sudo tee /var/www/html/index.html
sudo systemctl start httpd
sudo systemctl enable httpd
```

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3. Create a Target Group

- Navigate to the **AWS EC2 Console** → **Target Groups**.
- Click **Create Target Group**.
- Select **Target type:** `Instances`.
- Enter **Target Group Name**: `web-servers`.
- Choose **Protocol:** `HTTP` and **Port:** `80`.
- Select the **VPC** associated with your EC2 instances.
- Click **Next**.
- **Register Targets:**
  - Select `Server1` and `Server2`, then click **Include as pending**.
  - Click **Create Target Group**.

4. Create an Application Load Balancer (ALB)

- Navigate to **AWS EC2 Console** → **Load Balancers**.
- Click **Create Load Balancer** → **Application Load Balancer**.
- Configure the ALB:
  - **Name:** `web-load-balancer`.
  - **Scheme:** Internet-facing.
  - **Listeners:** HTTP (`80`).
  - **VPC:** Select your existing VPC.
  - **Subnets:** Select at least two subnets.
  - **Security Group:** Create or select an ALB-specific security group allowing HTTP (`80`) and HTTPS (`443`).
- **Configure Routing:**
  - Select the previously created target group `web-servers`.
- Click **Create Load Balancer**.

5. Test the Load Balancer

- Wait for the **ALB to become active**.
- Navigate to **AWS EC2 Console** → **Load Balancers**.
- Copy the **DNS Name** of the ALB.
- Open a web browser and **paste the DNS Name**.
- Refresh the page multiple times to verify that requests are being routed between `Server1` and `Server2`.

6. Cleanup Resources

- **Terminate EC2 instances** (`Server1` and `Server2`).
- **Delete the ALB** and **target group**.
- **Remove the security groups** created for this lab.
- Verify that all resources are deleted to prevent unnecessary charges.

Completion Criteria

- Security group created and configured correctly
- Two EC2 instances running Apache web servers
- Application Load Balancer successfully routing traffic
- Load balancing verified through ALB DNS
- All resources cleaned up after the lab

Submission Guidelines

- Take **screenshots** of each major step:
  - Security group configuration.
  - EC2 instance creation and Apache installation.
  - Target group creation and registration of instances.
  - Load Balancer setup and DNS testing.
  - Successful page refresh showing traffic routing.
- Save screenshots into a **document (Google Docs, PDF, etc.)**.
- Upload to **Google Drive** and share the link.
- Ensure **instructor has access** before submission.

🚀 **Great job! You've successfully deployed an Elastic Load Balancer to route traffic across multiple instances!** 🎉

AI TUTOR ACTIONS

Generate a practice quiz Summarize this unit Ask about this unit

Complete unit